Objective

To implement and demonstrate ArrayStoreException in Java using try-catch by attempting to store an incompatible data type in a String array.

Steps for Execution

1.Create a Java project and package named javacore.

2.Create a file named ArrayStoreExample.java and enter the given code.

3.Create a String array and assign it to an Object array reference.

4.Store valid String values in the array.

5.Try to store an integer value in the String array.

6.Handle the exception using catch.

7.Compile and run the program.

8.Observe the output.

Algorithm

1.Start the program.

2.Create a String array with three elements.

3.Assign the String array to an Object array reference.

4.Store String values in the array.

5.Try to store an integer value.

6.Catch ArrayStoreException.

7.Display the error message.

8.Display "Program ended.".

9.Stop the program.

UML Class Diagram

```
┌──────────────────────────────┐
   │   ArrayStoreExample   │
   ├──────────────────────────────┤
```

```
│ + main(args : String[])      │
│           : void             │
└──────────────┬───────────────┘
               │
               │ uses
               ▼
┌──────────────────────────────┐
│     ArrayStoreException      │
├──────────────────────────────┤
│ + Exception handling         │
└──────────────────────────────┘
```

Relationship

- ArrayStoreExample contains the main() method.
- The program handles ArrayStoreException using a try-catch block.
- The exception occurs because an Integer value (100) is attempted to be stored in a String array.

CODE:

```
ackage javacore;

public class ArrayStoreExample {

    public static void main(String[] args) {


        String[] names = new String[3];


        Object[] values = names;


        try {
            values[0] = "Ravi";
            values[1] = "Rahul";
```

```java
            System.out.println("Names added successfully.");

            values[2] = 100;
        }
        catch (ArrayStoreException e) {
            System.out.println("Error: Wrong type of value stored in array.");
        }

        System.out.println("Program ended.");
    }
}
```

Output

Names added successfully.

Error: Wrong type of value stored in array.

Program ended.

Result

The program successfully demonstrates ArrayStoreException handling in Java. A String array can store only String values, so attempting to store the integer 100 causes an ArrayStoreException, which is handled using the catch block.